11. Sequences and Series

11.2. Sequences

11.2b Learning Objectives

- Understand what a series means in relation to sequences
- Identify types of series (geometric and harmonic)
- Determine when a series converges or diverges

Theorem 11.2.1 **Divergence Theorem (Test for Divergence)**

If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$.

Therefore,

- If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- If $\lim_{n\to\infty} a_n = 0$, then the series may converge or diverge. The test is inconclusive.

Example 11.2.1

Use the divergence test to determine the divergence of the following series if applicable.

$$\sum_{k=1}^{\infty} \frac{k^2}{k^2 - 2k + 5}$$

Solution

Looking at the limit of the terms we have

$$\lim_{k\to\infty} \frac{k^2}{k^2 - 2k + 5} \overset{\text{LH}}{=} \lim_{k\to\infty} \frac{2k}{2k - 2} \overset{\text{LH}}{=} \lim_{k\to\infty} \frac{2}{2} = 1 \neq 0$$

Thus, by the Divergence Theorem, the series diverges.

Example 11.2.2

Use the divergence test to determine the divergence of the following series if applicable.

$$\sum_{n=1}^{\infty} \ln\left(\frac{n^2 + 1}{2n^2 + 1}\right)$$

Solution

Looking at the limit of the terms we have

$$\begin{aligned}\lim_{n\to\infty} \ln\left(\frac{n^2+1}{2n^2+1}\right) &= \ln\left(\lim_{n\to\infty} \frac{n^2+1}{2n^2+1}\right) \\ &\overset{\text{LH}}{=} \ln\left(\lim_{n\to\infty} \frac{2n}{4n}\right) \\ &= \ln\left(\frac{1}{2}\right) \neq 0\end{aligned}$$

Thus, by the Divergence Theorem, the series diverges.

Example 11.2.3

Use the divergence test to determine the divergence of the following series if applicable.

$$\sum_{n=1}^{\infty} \frac{1}{1+n}$$

Solution

Looking at the limit of the terms we have

$$\lim_{n\to\infty} \frac{1}{1+n} = 0$$

So the Divergence Theorem is inconclusive. (We cannot say if it diverges or converges yet).

Definition 11.2.1 Geometric Series

Geometric Series are series of the form

$$a + ar + ar^2 + \ldots + ar^{n-1} + \ldots = \sum_{n=1}^{\infty} ar^{n-1}$$

in which a and r are fixed real numbers and $a \neq 0$. The series can also be written

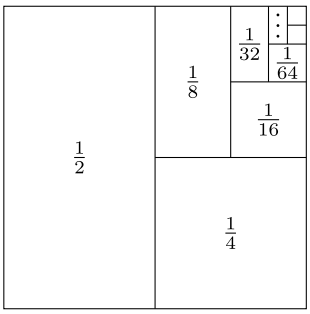
$$\sum_{n=0}^{\infty} ar^n$$

The ratio r can be positive or negative.

Remark

Geometric series are named this way because they often relate to proportions in geometric shapes. For example, consider $\sum_{n=1}^{\infty} \frac{1}{2^n}$.

$$\sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \ldots$$



In the limit, this series is equal to 1, which consists of the area of the entire square.

For any geometric series, we can determine the exact value if it converges.

$$\begin{aligned}
\sum_{n=1}^{\infty} ar^{n-1} &= a + ar + ar^2 + ar^3 + \dots \\
s_n &= a + ar + ar^2 + \dots + ar^{n-1} \\
rs_n &= ar + ar^2 + \dots + ar^{n-1} + ar^n \\
s_n - rs_n &= a - ar^n \\
s_n(1-r) &= a(1-r^n) \\
s_n &= \frac{a(1-r^n)}{1-r}
\end{aligned}$$

This sequence of partial sums converges precisely when $|r| < 1$ otherwise, $1 - r^n \to -\infty$. If $|r| < 1$, then $1 - r^n \to 1$, so

$$\sum_{n=1}^{\infty} = \frac{a}{1-r}.$$

Theorem 11.2.2

If $|r| < 1$, the **geometric series** $\sum_{n=1}^{\infty} ar^{n-1}$ converges, and

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$$

r is called the **ratio**

If $|r| \geq 1$ the series diverges.

Example 11.2.4

Find $\sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^k$.

Solution

We can rewrite this a little to look like

$$\sum_{k=1}^{\infty}\left(\frac{2}{3}\right)^{k}=\sum_{k=1}^{\infty}\frac{2}{3}\left(\frac{2}{3}\right)^{k-1}.$$

Then it is easy to see this is a geometric series with $r=\frac{2}{3}<1$ and $a=\frac{2}{3}$. So, this series converges to

$$\sum_{k=1}^{\infty}\frac{2}{3}\left(\frac{2}{3}\right)^{k-1}=\frac{a}{1-r}=\frac{\frac{2}{3}}{1-\frac{2}{3}}=2.$$

Example 11.2.5

Does the series $\sum_{n=1}^{\infty}2^{2n}5^{1-n}$ converge or diverge? If the series converges, then find its sum.

Solution

We rewrite the series with exponent rules to clearly see a and r

$$\begin{aligned}\sum_{n=1}^{\infty}2^{2n}5^{1-n}&=\sum_{n=1}^{\infty}\left(2^{2}\right)^{n}5^{1}5^{-n}\\&=\sum_{n=1}^{\infty}5^{1}(4)^{n}\left(5^{-1}\right)^{n}\\&=\sum_{n=1}^{\infty}5\left(\frac{4}{5}\right)^{n}\\&=\sum_{n=1}^{\infty}5\left(\frac{4}{5}\right)\left(\frac{4}{5}\right)^{n-1}\\&=\sum_{n=1}^{\infty}4\left(\frac{4}{5}\right)^{n-1}\end{aligned}$$

So this is a geometric series with $a=4$ and $r=\frac{4}{5}<1$. Because $r<1$, this series converges. So, it converges to

$$\sum_{n=1}^{\infty}4\left(\frac{4}{5}\right)^{n-1}=\frac{4}{1-\frac{4}{5}}=20.$$

Theorem 11.2.3 Algebraic Properties of Infinite Series

Given convergent series $\sum_{n=1}^{\infty}a_n$, $\sum_{n=1}^{\infty}b_n$, and any constant c, the following are true:

1. The series $\sum_{n=1}^{\infty}a_n+\sum_{n=1}^{\infty}b_n$
2. The series $\sum_{n=1}^{\infty}ca_n=c\sum_{n=1}^{\infty}a_n$

Example 11.2.6

Determine the convergence of the series

$$\sum_{n=1}^{\infty} \frac{2^n + e^n}{\pi^n}$$

Solution

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{2^n + e^n}{\pi^n} &= \sum_{n=1}^{\infty} \frac{2^n}{\pi^n} + \sum_{n=1}^{\infty} \frac{e^n}{\pi^n} \\ &= \sum_{n=1}^{\infty} \frac{2}{\pi}\left(\frac{2}{\pi}\right)^{n-1} + \sum_{n=1}^{\infty} \frac{e}{\pi}\left(\frac{e}{\pi}\right)^{n-1}\end{aligned}$$

Since these geometric series have $r = \frac{2}{\pi} < 1$ and $r = \frac{e}{\pi} < 1$ they converge. So

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{2^n + e^n}{\pi^n} &= \frac{\frac{2}{\pi}}{1 - \frac{2}{\pi}} + \frac{\frac{e}{\pi}}{1 - \frac{e}{\pi}} \\ &= \frac{2}{\pi - 2} + \frac{e}{\pi - e}\end{aligned}$$

The series converges.

Example 11.2.7

Find the values of x for which the series converges.

$$\sum_{n=0}^{\infty} x^n$$

Solution

This is a geometric series for all x, so it converges when $|x| < 1$. This is because x is the ratio, so this converges for x in $(-1, 1)$.

This is called an interval of convergence, which we will discuss more later.

Example 11.2.8

Find the values of x for which the series converges.

$$\sum_{n=0}^{\infty} (-4)^{n(x-5)^n}$$

Solution

$$\sum_{n=0}^{\infty} (-4)^{n(x-5)^n} = \sum_{n=0}^{\infty} (-4(x-5))^n$$

This converges when $r = -4(x-5)$ has $|r| < 1$, so

$$\begin{aligned} |-4(x-5)| &< 1 \\ 4\,|x-5| &< 1 \\ |x-5| &< \frac{1}{4} \end{aligned}$$

So $x - 5 < \frac{1}{4} \Rightarrow x < \frac{21}{4}$ and $x - 5 > -\frac{1}{4} \Rightarrow x > \frac{19}{4}$. So the interval of x where this series converges is $\left(\frac{19}{4}, \frac{21}{4}\right)$.

Example 11.2.9 **Extra Example**

Malaria is a parasitic infection transmitted by mosquito bites, mainly in tropical area of the world. Suppose a person is given a 50 mg dose of quinine at the same time every day for the prevention of malaria. After the first dose, the person has 50 mg of quinine in the body. Each day, the person's body metabolizes some of the quinine so that, after one day 23% of the original amount remains. How much quinine remains in the body after a long time with a dose every day?

Solution

We need to add up the total amount of quinine, so we will track how much is left after each day.
Day 1: 50

Day 2: $50 + (0.23)50$

Day 3: $50 + (0.23)50 + (0.23)^2 50$

Day 4: $50 + (0.23)50 + (0.23)^2 50 + (0.23)^3 50$

and so on. Thus, the total amount after a very long time is

$$\sum_{n=1}^{\infty} 50(0.23)^n - 1 = \frac{50}{1-0.23} = \frac{50}{0.77} \approx 64.95\text{mg}$$

11.2b Section Summary:

- We used the divergence test to tell if series diverged.
- We went over geometric series and the formula for their value.
- We went over algebraic properties of series.